

Team B

Case Summary: Worldwide Paper Company – Longwood Woodyard Investment

Bob Prescott, the controller at Blue Ridge Mill, is considering a \$18 million investment to build a new woodyard on-site. This woodyard would allow the mill to process whole tree logs (called longwood) directly instead of buying shorter wood pieces (shortwood) from a nearby competitor, Shenandoah Mill.

By making this change, Blue Ridge Mill expects to save money by no longer purchasing shortwood—and potentially make extra revenue by selling any leftover shortwood on the open market.

1. Project Overview

- Build a new woodyard to process longwood logs.
- Replace reliance on competitor's shortwood supply.
- Benefits: cost savings + new revenue from selling excess shortwood.

2. Investment Details

- Total cost: \$18 million
- Spending timeline: \$16 million in 2016, \$2 million in 2017.

3. Expected Financial Impact

- Cost savings: \$2 million in 2017, \$3.5 million annually thereafter.
- New sales revenue: \$4 million in 2017, growing to \$10 million annually from 2018–2022.
- Costs:
 - Cost of goods sold (COGS): 75% of revenue.
 - Selling & administrative expenses (SG&A): 5% of revenue.

4. Working Capital Needs

- Additional working capital equals 10% of incremental revenues each year.
- Working capital fully recovered by end of 2022.
- Only 10% (\$1.8 million) of the initial capital investment is recoverable at project end.

5. Accounting & Tax Assumptions

- Depreciation: straight-line over 6 years, starting in 2017, no salvage value.
- Tax rate: 40%.
- Inflation ignored beyond 2017, assuming cost increases offset by revenue increases.

6. Discount Rate & Company Financials

- Company hurdle rate (discount rate): 10% but based on data from 10 years ago.
- The company has a strong financial position and a solid credit rating (A).

7. Key Decision

- Is the \$18 million investment worth it?
- Should Prescott approve the project based on net present value (NPV) using the company's hurdle rate?

Debt and equity value inputs

What are reasonable inputs you could use for the debt and equity value inputs needed to calculate WPC's weighted average cost of capital (i.e., D and E)?

Input	Value
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Shares Outstanding	\$500,000,000.00
Market Price per Share	\$24.00
Market Value of Equity (E)	\$12,000,000,000.00
Bank Loan Payable	\$500,000,000.00
Long-Term Debt	\$2,500,000,000.00
Total Debt (D)	\$3,000,000,000.00
Total Capital (E + D)	\$15,000,000,000.00
% Equity	80.00%
% Debt	20.00%

Cost of debt Input

What is a reasonable cost of debt input for WPC's weighted average cost of capital? Please write 1-2 sentences to explain your choice here.

A reasonable cost of debt for WPC is 3.85%, based on the yield of 10-year A-rated corporate bonds shown in Exhibit 1. This is appropriate because WPC has an A credit rating, and 10 years aligns with the likely term of the investment.

Input	Value	Formula / Notes
Bond Rating	A	Given in case (WPC has an A rating)

10-Year A Bond Yield	3.85%	From Exhibit 1: yield on 10-year A-rated corporate bonds
Cost of Debt (rd)	3.85%	Use 3.85% as reasonable estimate
Explanation	Based on 10-year A-rated bond yields, which match WPC's credit rating and maturity profile.	

Cost of equity input

What is a reasonable cost of equity input for WPC's weighted average cost of capital? Please briefly explain how you arrived at your answer and found the necessary input.

Input	Value
Risk-Free Rate (rf)	2.04%
Beta (β)	1.1

Market Risk Premium (MRP)	6.00%
Cost of Equity (re)	8.64%

I calculated the cost of equity using the Capital Asset Pricing Model (CAPM), which requires three inputs: the risk-free rate, beta, and market risk premium. I used the 10-year government bond yield of 2.04% as the risk-free rate because it reflects the return on a long-term, low-risk investment. WPC's beta of 1.10 measures its stock volatility relative to the market, and the 6.0% market risk premium represents the historical excess return investors expect from the market.

WACC and discussion of hurdle rate

The case says WPC uses a hurdle rate of 10% when calculating the NPV of its projects. Do you think that is reasonable? Why or why not? And if not, do you think this company will tend to accept too few or too many projects if it continues to use a hurdle rate of 10% when calculating the NPV?

Topic	Value / Explanation
Current Hurdle Rate	10%
Reasonableness of Hurdle	The 10% rate is outdated, based on a study from 10 years ago when Treasury yields were ~4.7%
Updated Market Rates	Current 10-year Treasury yield is ~2%, and WPC's WACC is likely below 10%
Implication of High Rate	Using 10% as a hurdle means WPC may reject projects with returns between WACC and 10%
Effect on Project Acceptance	WPC will accept too few projects, missing profitable investments and limiting growth

The 10% hurdle rate WPC uses is likely unreasonable given current market conditions. It was set based on a cost of capital study from 10 years ago when interest rates were much higher (with 30-year Treasury bonds yielding about 4.7%). Now, long-term government bond yields are closer to 2%, and with WPC's current cost of debt and equity, the actual weighted average cost of capital (WACC) is probably well below 10%.

If WPC continues to use the outdated 10% hurdle rate, it will tend to reject projects that have positive net present value but whose returns fall between the true WACC and 10%. This means the company will accept too few projects, potentially missing out on value-creating investments and limiting growth.

Additional discussion

If the proposed project is considerably riskier than the company's existing operations, how would that affect your assessment on whether the weighted cost of capital you calculated is an appropriate discount rate for this project?

If the proposed project is significantly riskier than WPC's current operations, then the company's overall weighted average cost of capital (WACC) would understate the true risk of the project. In this case, using the company's average WACC as the discount rate would be inappropriate, because it assumes the project has the same risk profile as the firm's typical investments.

For riskier projects, a higher discount rate should be used to reflect the additional risk premium investors require. This means you would adjust the discount rate upward, either by increasing the beta (to capture higher systematic risk) or by adding a risk premium on top of the standard WACC. This adjustment helps avoid overvaluing the project and ensures the NPV calculation properly accounts for the higher uncertainty and potential variability in returns.

